



The Second-Order Theory

- The Second-Order Linear ODEs

- The Standard Homogeneous 2nd-Order

- Linear ODE with Constant Coefficients

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = 0 \quad \text{---} \quad (*)$$

where $a, b, c \in \mathbb{R}$.

$$\rightarrow \frac{d^2 y}{dx^2} + p \frac{dy}{dx} + qy = 0 \quad \text{---} \quad \tilde{(*)}$$

where $p, q \in \mathbb{R}$.

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(a) $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 3y = e^{-2x}$ non-constant
Non Homog (b) $\frac{d^2y}{dx^2} + 2y = 0$

(c) $\frac{d^2x}{dt^2} + 3\frac{dx}{dt} + 7x = 0$ (d) $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 0$

(c) & (d) are constant coefficient
 homogeneous 2nd order ODE.

Homogeneous Linear ODEs: Superposition of Solutions

The functions $y = \cos x$ and $y = \sin x$ are solutions of the homogeneous linear ODE

$$y'' + y = 0$$

for all x . We verify this by differentiation and substitution. We obtain $(\cos x)'' = -\cos x$; hence

$$y'' + y = (\cos x)'' + \cos x = -\cos x + \cos x = 0.$$

Similarly for $y = \sin x$ (verify!). We can go an important step further. We multiply $\cos x$ by any constant, for instance, 4.7, and $\sin x$ by, say, -2, and take the sum of the results, claiming that it is a solution. Indeed, differentiation and substitution gives

$$(4.7 \cos x - 2 \sin x)'' + (4.7 \cos x - 2 \sin x) = -4.7 \cos x + 2 \sin x + 4.7 \cos x - 2 \sin x = 0. \quad \blacksquare$$

In this example we have obtained from $y_1 (= \cos x)$ and $y_2 (= \sin x)$ a function of the form

$$(3) \quad y = c_1 y_1 + c_2 y_2 \quad (c_1, c_2 \text{ arbitrary constants}).$$

This is called a linear combination of y_1 and y_2 . In terms of this concept we can now formulate the result suggested by our example, often called the superposition principle or linearity principle.

Important concept for the solution
 of 2nd order ODE.

THEOREM

Fundamental Theorem for the Homogeneous Linear ODE (2)

For a homogeneous linear ODE (2), any linear combination of two solutions on an open interval I is again a solution of (2) on I . In particular, for such an equation, sums and constant multiples of solutions are again solutions.

PROOF Let y_1 and y_2 be solutions of (2) on I . Then by substituting $y = c_1y_1 + c_2y_2$ and its derivatives into (2), and using the familiar rule $(c_1y_1 + c_2y_2)' = c_1y_1' + c_2y_2'$, etc., we get

$$\begin{aligned} y'' + py' + qy &= (c_1y_1 + c_2y_2)'' + p(c_1y_1 + c_2y_2)' + q(c_1y_1 + c_2y_2) \\ &= c_1y_1'' + c_2y_2'' + p(c_1y_1' + c_2y_2') + q(c_1y_1 + c_2y_2) \\ &= c_1(y_1'' + py_1' + qy_1) + c_2(y_2'' + py_2' + qy_2) = 0, \end{aligned}$$

since in the last line, $(\dots) = 0$ because y_1 and y_2 are solutions, by assumption. This shows that y is a solution of (2) on I . ■

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Solution Procedure

Assumption

Euler's Substitution: Let $y = e^{\lambda x}$.

$$\frac{dy}{dx} = \lambda e^{\lambda x} \text{ and } \frac{d^2y}{dx^2} = \lambda^2 e^{\lambda x}.$$

This implies that $\lambda^2 e^{\lambda x} + p\lambda e^{\lambda x} + qe^{\lambda x} = 0$.

Since, $e^{\lambda x} \neq 0$, therefore,

$$\boxed{\lambda^2 + p\lambda + q = 0} \text{ --- } \tilde{f}$$

$\tilde{f}$ is called the characteristic (or auxiliary) equation for $\tilde{x}$. [Remark: The values of λ 's are said to be the characteristic roots of $\tilde{f}$ (or $\tilde{x}$).

Reason:

It isn't just second order linear ODEs with constant coefficients. All linear ODEs with constant coefficients have solutions that are linear combinations of exponentials, and polynomials multiplied by exponentials.

This is due to the fact that the exponential function is an eigenfunction of the derivative operator.

Key point.

$$\frac{d}{dx} e^{\lambda x} = \lambda e^{\lambda x}$$

A linear ODE with constant coefficients can be written as a differential operator acting on a

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A linear ODE with constant coefficients can be written as a differential operator acting on a function. Then since we know what eigenfunction works for each of its parts it turns into solving the characteristic equation. Knowing the eigenfunctions allows us to convert an ODE into an algebra problem.

Example:

$$y'' + 5y' + 6y = 0$$

$$\left(\frac{d^2}{dx^2} + 5 \frac{d}{dx} + 6 \right) y = 0$$

When y is decomposed into exponentials each derivative just pops out an eigenvalue.

Case No.1: (if $p^2 - 4q > 0$) Two Real Distinct Roots

$$\lambda_1 = -\frac{p}{2} - \frac{\sqrt{p^2 - 4q}}{2}$$

$$\lambda_2 = -\frac{p}{2} + \frac{\sqrt{p^2 - 4q}}{2}$$

Two Basic Solutions of $\tilde{y}$:

$$y_1 = e^{\lambda_1 x} \text{ and } y_2 = e^{\lambda_2 x}$$

The complete solution of $\tilde{y}$ is: $y = C_1 e^{\lambda_1 x} + C_2 e^{\lambda_2 x}$, where C_1 and C_2 are two arbitrary constants.

• These two solutions are linearly independent i.e. if $C_1 e^{\lambda_1 x} + C_2 e^{\lambda_2 x} = 0, \forall x \in \mathbb{R}$, then $C_1 = C_2 = 0$.

$$\frac{d^2 y}{dx^2} + 5 \frac{dy}{dx} + 6y = 0 \quad \text{--- } \text{if}$$

$$\lambda^2 + 5\lambda + 6 = 0 \quad \leftarrow \text{Characteristic Eq}$$

$$\lambda = -3, -2$$

$$y = C_1 y_1 + C_2 y_2 = C_1 e^{\lambda_1 x} + C_2 e^{\lambda_2 x}$$

$$y(x) = C_1 e^{-3x} + C_2 e^{-2x}$$

Solution

Case No. 2: ($p^2 - 4q = 0$) Two Real Repeated Roots

$$\text{Let } \lambda = \lambda_1 = \lambda_2 = -b/2.$$

Two Basis Independent solutions:

$$y_1 = e^{\lambda x} \text{ and } y_2 = x e^{\lambda x}$$

The solution of $\tilde{F}$: $y = (C_1 + C_2 x) e^{\lambda x}$

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The solution of $\tilde{F}$: $y = (C_1 + C_2 x) e^{\lambda x}$

$$\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + y = 0 \quad \text{--- } \tilde{F}$$

$$\lambda^2 - 2\lambda + 1 = 0$$

$$(\lambda - 1)^2 = 0$$

$$\lambda = 1, 1$$

$$y(x) = C_1 e^x + C_2 x e^x$$

Solution

Case No. 3: ($p^2 - 4q < 0$): No real roots/Complex Roots:

$$\lambda_1 = -\frac{p}{2} - \frac{\sqrt{4q - p^2}}{2} i \text{ and}$$

$$\lambda_2 = -\frac{p}{2} + \frac{\sqrt{4q - p^2}}{2} i.$$

$$\text{Let } \alpha = p/2, \beta = \frac{\sqrt{4q - p^2}}{2}.$$

So our roots are: $\lambda_1 = -\alpha - \beta i$ and $\lambda_2 = -\alpha + \beta i$.

Version No 1:

$$y = e^{-\alpha x} [C_1 e^{-\beta x i} + C_2 e^{\beta x i}]$$

Version No 2:

$$\text{Ex: } y = e^{-\alpha x} [C_1 \cos \beta x + C_2 \sin \beta x]$$

[Hint: Euler's Famous Identity: $e^{i\theta} = \cos \theta + i \sin \theta$.]

Any form

Version No 3:

$$\text{Ex: } y = e^{-\alpha x} C_1 \sin(\beta x + C_2) = e^{-\alpha x} R \sin(\beta x + \phi).$$

(Ex) $y = e^{C_1 \sin(\beta x + C_2)} = e^{\overline{R} \sin(\beta x + \phi)}$.
[Hint: See Harmonic Forms.]

$$\frac{d^2 y}{dx^2} + \frac{dy}{dx} + y = 0$$

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$$\lambda^2 + \lambda + 1 = 0$$

$$\alpha = \frac{1}{2} \quad \& \quad \beta = \frac{\sqrt{3}}{2}$$

$$y(x) = e^{\frac{-x}{2}} \left(C_1 \cos\left(\frac{\sqrt{3}}{2} x\right) + C_2 \sin\left(\frac{\sqrt{3}}{2} x\right) \right)$$

Solution

Summary of Cases I–III

Case	Roots of (2)	Basis of (1)	General Solution of (1)
I	Distinct real λ_1, λ_2	$e^{\lambda_1 x}, e^{\lambda_2 x}$	$y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$
II	Real double root $\lambda = -\frac{1}{2}a$	$e^{-ax/2}, x e^{-ax/2}$	$y = (c_1 + c_2 x) e^{-ax/2}$
III	Complex conjugate $\lambda_1 = -\frac{1}{2}a + i\omega,$ $\lambda_2 = -\frac{1}{2}a - i\omega$	$e^{-ax/2} \cos \omega x$ $e^{-ax/2} \sin \omega x$	$y = e^{-ax/2} (A \cos \omega x + B \sin \omega x)$

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$$e^z = e^{r+it} = e^r e^{it} = e^r (\cos t + i \sin t).$$

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$$e^{it} = \cos t + i \sin t,$$

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